UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Unit 1 · Partial Differentiation · all sessions (Class 001 – Class 011) · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S001·15	Deck 001, slide 15 (rendered .pptx)
T001a 00:14:20	Transcript "Introduction to Partial Differentiation", segment start
T001b 00:11:41	Transcript "Continuity and differentiability of a function of two independent variables"
CW U1 P7	Unit 1 class-work problem 7
ESA Dec-23 1a	Past paper: Dec 2023, question 1.a
S002·7	Deck 002, slide 7 (rendered .pptx)
T002 00:06:10	AV-summary transcript 002, segment start time
V002 00:07:45	Frame grabbed from the class 002 video
CW U1 P2 · ESA Dec-23 1a	Unit 1 class-work problem 2 · past-paper question
S003·7 · S004·9	Deck 003 / 004, slide number (rendered .pptx)
T003a 00:06:27	Transcript 003 part I: "Total derivative, chain rule and implicit functions"
T003b 00:01:10	Transcript 003 part II: "Problems on total derivatives, chain rule"
T004a 00:13:09	Transcript 004 part I: "Implicit functions and problems"
T004b 00:02:31	Transcript 004 part II: "Problems on composite and implicit functions"
CW U1 P14 · ESA Jan-24 s3 1b	Unit 1 class-work problem 14 · past-paper question
S005·6	Deck 005, slide 6 (rendered .pptx)
T005 00:08:57	AV-summary transcript 005, segment start
CW U1 P20	Unit 1 class-work problem 20
ESA Jan-24 s3 1c	Past paper: Jan 2024, set 3, question 1.c
S006·12	Deck 006, slide 12 (rendered .pptx)
T006a 00:05:18	Transcript 006 part I (slides 3–4, 9.1 min)
T006b 00:08:20	Transcript 006 part II (slides 5–17, 12.9 min)

S007·9	Deck 007, slide 9 (rendered .pptx)
T007 00:18:07	AV-summary transcript 007, segment start
CW U1 P24	Unit 1 class-work problem 24
ESA Dec-23 1b	Past paper: Dec 2023, question 1.b
S008·12	Deck 008, slide 12 (rendered .pptx)
T008 00:07:38	AV-summary transcript 008, segment start
S009·7	Deck 009, slide 7 (rendered .pptx)
T009 00:07:04	AV-summary transcript 009 (“Extremas of functions”), segment start
CW U1 P31	Unit 1 class-work problem 31
ESA Dec-23 1c	Past paper: Dec 2023, question 1.c
S010·8	Deck 010, slide 8 (rendered .pptx)
T010a 00:07:38	AV-summary transcript “Extremas of functions continued” (older lecture), segment start
T010b 00:03:12	AV-summary transcript “Problems on Maxima and Minima for a function of two variables” (newer lecture)
CW U1 P29	Unit 1 class-work problem 29
ESA Jan-24 s2 1a	Past paper: Jan 2024 set 2, question 1.a
S011·5	Deck 011, slide 5 (rendered .pptx)
T011a 00:01:08	Transcript “Lagrange's method of undetermined coefficient” (theory lecture), segment start
T011b 00:05:03	Transcript “Problems on Lagrange's method of undetermined multipliers”
CW U1 P33	Unit 1 class-work problem 33
Extra Q57	Extra question list “lagrange-ques”, question 57
ESA Jul-23 1c	Past paper: Jul 2023, question 1.c

Contents

1. 1 · Functions of two or more variables
2. 2 · Limit of a function of two variables
3. 3 · Continuity of a function of two variables
4. 4 · Partial derivatives: history, definition and notation
5. 5 · Second-order and mixed partial derivatives
6. 6 · Higher-order partial derivatives
7. 7 · The one-variable picture: derivative as slope
8. 8 · A function of two variables is a surface
9. 9 · First-order partials as slopes of tangents
10. 10 · Slope in any other direction: the gradient
11. 11 · Second-order partials: concavity and mixed partials
12. 12 · Problems on partial derivatives
13. 13 · Composite functions and the one-variable chain rule

14. 14 · The total derivative rule
15. 15 · The total differential: approximation and sensitivity
16. 16 · Rates of change problems
17. 17 · The chain rule for composite functions
18. 18 · Implicit functions of two variables
19. 19 · Implicit functions of three variables
20. 20 · Homogeneous functions
21. 21 · Euler's theorem
22. 22 · Second and higher order: the corollary
23. 23 · Extension: when f(u) is homogeneous
24. 24 · Applications of homogeneous functions
25. 25 · Testing homogeneity: the scaling test
26. 26 · Verifying Euler's theorem directly
27. 27 · When u itself is homogeneous: first and second order
28. 28 · When f(u) is homogeneous: the extension
29. 29 · Why Taylor's and Maclaurin's series?
30. 30 · Taylor's and Maclaurin's series in one variable
31. 31 · Taylor's expansion of a function of two variables
32. 32 · Maclaurin's expansion of a function of two variables
33. 33 · Taylor expansion about a point (a, b)
34. 34 · Maclaurin expansion about the origin
35. 35 · Approximate values from a Taylor polynomial
36. 36 · Maxima, minima and saddle points
37. 37 · Definitions: relative maximum, minimum and extreme value
38. 38 · Conditions and the working rule
39. 39 · The method, and the ESA cubic functions
40. 40 · Functions with many stationary points
41. 41 · Applied problems: eliminate one variable first
42. 42 · Lagrange's method: the auxiliary function and the procedure
43. 43 · Geometric problems: inscribed solids, nearest and farthest points
44. 44 · Applied problems: boxes, splitting a number, temperature
45. Past-paper questions

M1

Class 001 · Introduction to Partial Differentiation

1 Functions of two or more variables

Definitions

Function of two variables. A function of two variables denoted by $z = f(x, y)$ is a rule that assigns to each ordered pair of real numbers (x, y) in a set D a unique real number denoted by $f(x, y)$. The set D is the domain of f and its range is the set of values that f takes on: $\{f(x, y) : (x, y) \in D\}$. S001·8 · T001a 00:05:46

Independent and dependent variables. In $z = f(x, y)$, x and y are independent variables and z is the dependent variable (a bivariate function); with three independent variables one writes $w = f(x, y, z)$, and in general $f(x_1, x_2, \ldots, x_n)$. (transcript only) T001a 00:02:18 · T001a 00:02:54

Key points

1. Examples on the slides: area of a rectangle $A(l, b)$; volume of a cylinder $v(r, h)$; electric potential in a cathode-ray tube $V = f(x, y, z, t)$; propagation of an electromagnetic wave (magnetic field, electric field, wavelength, direction); weather (topography, pressure, humidity, temperature); spread of a pandemic (area, population, awareness, immunity, age and gender distribution). ESA ★ S001·4 · S001·5 · S001·6 · S001·7 · T001a 00:03:26
2. A single-variable function is enough to understand a concept, but to draw a conclusion about a real situation we almost always need several variables. ESA ★ (transcript only) T001a 00:05:16
3. f maps an ordered pair to one real number: for $z = x^3y$, the pair $(2, 1)$ is mapped to 8. ESA ★ T001a 00:07:16
4. Graph of $z = f(x, y)$: a surface, which needs three axes; a function of three variables would need four dimensions, so the unit sticks to two variables. The slide shows $\sin x \cos y$ and $e^{1-x^2-y^2}$ (Figures G1.1–G1.2). ESA ★ S001·8 · T001a 00:08:35 · T001a 00:09:07

2 Limit of a function of two variables

Definitions

Limit of a function of two variables. A function $f(x, y)$ is said to approach the limit L as (x, y) approaches (a, b), if the limit remains the same along any path of (x, y) approaching (a, b): $\lim_{(x,y)\to(a,b)} f(x, y) = L$. S001·9 · T001a 00:11:01

Key points

1. One variable: $x \to a$ from the left or the right only. Two variables: $(x, y) \to (a, b)$ along any of infinitely many paths (Figure G2.1). ESA ★ S001·9 · T001a 00:11:32 · T001a 00:12:34
2. If substitution gives a number (a polynomial, no $\frac{0}{0}$), that number is the limit, whatever the path. ESA ★ T001a 00:13:59
3. It is hard to show that a limit exists but easy to show it does not: two paths with different values are enough. ESA ★★ S001·10 · T001a 00:17:42
4. Agreement along a few paths proves nothing: for $\frac{2xy}{3x^2+y^2}$, the x-axis and the parabola $y = x^2$ both give 0, but $y = x$ gives $\frac{1}{2}$. ESA ★★ T001a 00:16:48 · T001a 00:17:24

Why?

Why can one path never prove that a limit exists?
There are infinitely many paths, and a function can agree on any finite number of them and still differ on another: $\frac{2xy}{3x^2+y^2}$ gives 0 on the x-axis and on $y = x^2$ but $\frac{1}{2}$ on $y = x$. To prove existence one needs an estimate for all paths at once (e.g. polar coordinates: $\frac{3x^2y}{x^2+y^2} = 3r\cos^2\theta\sin\theta \to 0$).
ADDED FOR CLARITY T001a 00:17:42 · S001·12

Exam traps

Exam trap. Same value along two or three paths does not prove that the limit exists; different values along two paths does prove it does not. T001a 00:16:48 · T001a 00:17:42

Exam trap. Slide 001·9 uses the point $(2, 1)$ (limit -8); the lecture uses $(2, -1)$ (limit -6). Read the point in the question. ADDED FOR CLARITY S001·9 · T001a 00:12:59

3 Continuity of a function of two variables

Definitions

Continuity at a point. $f(x,y)$ is continuous at (a,b) if $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$: $f(a,b)$ must be defined, the limit must exist, and the two must be equal. S001·11 · T001b 00:00:38 · T001a 00:18:46

Statements in the class material

> *"A function is continuous at a point if the graph doesn't have any holes or breaks at that point. Graphing the function is helpful. If a function is not continuous at (a,b), this does not mean the limit does not exist. We will need to consider the approaching (a,b) from different paths."* S001·11

Key points

1. One variable: f is continuous at c if (i) $\lim_{x\to c} f(x)$ exists, (ii) $f(c)$ exists, (iii) they are equal. The same three conditions define continuity for $f(x,y)$. ESA ★ T001a 00:18:46 · S001·11
2. $\frac{3x^2y}{x^2+y^2}$: the limit at the origin exists (it is 0) but $f(0,0)$ is $\frac{0}{0}$, so f is not continuous there. $\frac{2xy}{3x^2+y^2}$: the limit does not exist and $f(0,0)$ is undefined (Figures G3.1–G3.2). ESA ★ S001·12 · S001·13 · T001b 00:04:14 · T001b 00:05:44
3. To decide continuity in higher dimensions, graph the function: a hole, fold or sudden drop in the surface shows a discontinuity. ESA ★ S001·11 · T001b 00:06:45 · T001a 00:18:07

Why?

Why is $\frac{3x^2y}{x^2+y^2}$ not continuous at the origin even though its limit is 0?
Continuity needs $f(0,0)$ to be defined and equal to the limit. Here $f(0,0)$ is $\frac{0}{0}$, so the third condition cannot be checked. Defining $f(0,0)=0$ would make it continuous.
ADDED FOR CLARITY S001·12 · T001b 00:04:14

Exam traps

Exam trap. "Not continuous" does not mean "the limit does not exist" (slide 001·11): $\frac{3x^2y}{x^2+y^2}$ has limit 0 at the origin. S001·11 · T001b 00:02:10

Exam trap. Continuous is not the same as smooth: $z=|x|+|y|$ is continuous everywhere but has a corner at the origin. The lecture's "continuous if the surface is smooth" is loose.
ADDED FOR CLARITY T001b 00:06:26

4 Partial derivatives: history, definition and notation

Definitions

Partial derivative. A partial derivative of a function of several variables is the derivative with respect to one of those variables, with the others held constant. S001·15 · T001b 00:12:52

Key points

1. The product rule is already partial differentiation: in $\frac{d}{dx}(x^3 \sin x) = x^3 \cos x + 3x^2 \sin x$, each term treats the other factor as a constant k. Leibniz: $d(uv) = u\,dv + v\,du$. ESA ★ S001·14 · T001b 00:07:56 · T001b 00:10:18
2. History: Leibniz first used partial differentiation (product rule); Adrien-Marie Legendre created the modern symbol ∂ in 1786 but later abandoned it; Carl Gustav Jacob Jacobi reintroduced it in 1841. ESA ★ S001·14 · T001b 00:11:23
3. To find z_x, look only at the terms containing x; everything else is a constant, however many other variables there are. ESA ★★ T001b 00:15:08 · T001b 00:15:48
4. A function of x, y, t has three first partials z_x, z_y, z_t; for z_x treat y and t as constants. ESA ★ T001b 00:13:20
5. All rules of ordinary differentiation hold for partial derivatives: product, quotient, chain rule, implicit and composite functions. ESA ★★ T001b 00:13:57 · T001b 00:14:16

Why?

Why does every rule of ordinary differentiation still work?
With the other variables frozen, $f(x, y)$ is an ordinary function of x alone, and $\frac{\partial f}{\partial x}$ is its ordinary derivative (E4.1 is the usual limit with y fixed).

T001b 00:13:57 · S001·15

Exam traps

Exam trap. The partial symbol is ∂, not δ (the lecture stresses this) and not d. T001b 00:11:41

5 Second-order and mixed partial derivatives

Definitions

Mixed (crossed) partial derivatives. $z_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)$ and $z_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right)$: differentiate with respect to one variable, then the other. S001·16 · T001b 00:18:24

Statements in the class material

> *"Note 1: The crossed or mixed partial derivatives are in general equal (Clairaut's theorem):* $\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)$*. That is, the order of differentiation is immaterial if the second order derivatives involved are continuous."* S001·17 · T001b 00:19:57

> *"Note 2: In the subscript notation, the subscripts are written in same order in which the differentiation is carried out, while in the* ∂ *notation the order is opposite. For example,* $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = f_{xy}$*."* S001·17

Key points

1. Read z_{xyxx} left to right: $z \to z_x \to z_{xy} \to z_{xyx} \to z_{xyxx}$ (differentiate in x, then y, then x, then x). ESA ★ T001b 00:18:51 · T001b 00:19:21
2. Clairaut's theorem is "a very important theorem": whenever the second partials are continuous (true for all examination functions away from bad points), compute whichever mixed partial is easier. ESA ★★★ S001·17 · T001b 00:20:10
3. Verifying $f_{xy} = f_{yx}$ is a standard class-work question: x^y (CW P2) and $x^2 \tan^{-1}\frac{y}{x} - y^2 \tan^{-1}\frac{x}{y}$ (CW P8, W5.3). ESA ★★ CW U1 P2 · CW U1 P8

Exam traps

Exam trap. $\frac{\partial^2 z}{\partial y\, \partial x}$ means x first, then y: it is z_{xy}, not z_{yx}. S001·17

Exam trap. Clairaut needs continuous second partials: for $f = \frac{xy(x^2-y^2)}{x^2+y^2}$, $f(0,0) = 0$, one gets $f_{xy}(0,0) = -1$ but $f_{yx}(0,0) = 1$. (ADDED FOR CLARITY) S001·17

M2

Class 002 · Geometrical Interpretation of Partial Derivatives

6 Higher-order partial derivatives

Definitions

First-order partial derivative. The partial derivative $\partial f/\partial x$ obtained by differentiating once (here, with respect to x while y is held constant). S002·3

Second-order partial derivatives. $\dfrac{\partial^2 f}{\partial x^2}$, $\dfrac{\partial^2 f}{\partial y^2}$, $\dfrac{\partial^2 f}{\partial x \partial y}$, $\dfrac{\partial^2 f}{\partial y \partial x}$, which are obtained by differentiating twice. S002·3

Higher-order partial derivatives. Third, fourth and nth order derivatives involve 3, 4 or n times differentiation respectively. S002·3

Mixed partial derivative. A higher-order partial in which you differentiate with respect to more than one variable, e.g. f_{xy} and f_{yx}. S002·9 · T002 00:00:06

Statements in the class material

> *"A function of 2 variables has two first order derivatives, four second order derivatives and 2^n of nth order derivatives. A function of m independent variables will have m^n derivatives of order n."* S002·3

Key points

1. A partial derivative is taken with respect to one variable while every other variable is held constant. ESA ★★ T002 00:00:06
2. Differentiating a first-order partial again gives a second-order partial; n differentiations give an nth-order partial. ESA ★★ S002·3
3. For $f(x, y)$: 2 first-order (f_x, f_y), 4 second-order ($f_{xx}, f_{yy}, f_{xy}, f_{yx}$), 2^n of order n. ESA ★★ S002·3 · T002 00:00:44
4. For $f(x_1, \ldots, x_m)$: m^n derivatives of order n; e.g. m^2 second-order partials. ESA ★ S002·3 · T002 00:02:13
5. The two mixed partials "were equal" in the class examples: $f_{xy} = f_{yx}$ for the smooth functions of this unit. ESA ★★ (transcript only) T002 00:00:24

Exam traps

Exam trap. In f_{xy} the order is x first, then y. Some books write the same derivative as $\partial^2 f / \partial y \, \partial x$. Both mean the same, and for this unit's functions the order does not change the answer. ADDED FOR CLARITY

7 The one-variable picture: derivative as slope

Definitions

Derivative (one variable). For $y = f(x)$, dy/dx is the change in y produced by a change in position along the x-axis; at a point it equals the slope of the tangent to the curve. S002·4 · T002 00:02:50

Statements in the class material

> *"We know that the derivative of a single valued function gives the slope of the tangent line to the curve at the given point."* S002·4

Key points

1. A derivative always represents change: dy/dx is the change on the curve as you change your position on the x-axis. ESA ★ T002 00:02:50 · T002 00:02:59
2. At a particular point, the derivative is the slope of the tangent at that point. ESA ★ S002·4 · T002 00:03:13

8 A function of two variables is a surface

Definitions

Bivariate function. A function $z = f(x, y)$ of two independent variables. Its graph is a surface in three dimensions. T002 00:04:16

Curve on a surface. If the points (x, y) are restricted to a curve in the xy-plane (circle, parabola, ...), the points $(x, y, f(x, y))$ form a curve that lies on the surface. S002·5

Statements in the class material

> *"If the points (x,y) lie on a circle in the xy plane, then z = f(x,y) is a curve which looks like the graph shown. If the points (x,y) lie on a parabola in the xy plane, then z = f(x,y) is a curve which looks like the graph shown."* S002·5

Key points

1. A function of two independent variables always gives a surface in three dimensions: each (x, y) has one z. ESA ★ T002 00:04:16
2. Restricting (x, y) to a curve in the xy-plane (circle, parabola) picks out a curve on the surface. ESA ★ S002·5 · T002 00:03:32

9 First-order partials as slopes of tangents

Definitions

Partial derivative, geometrically. The slope of the tangent line to the surface $z = f(x, y)$ at (x_0, y_0), along the direction of the variable you differentiate with respect to. S002·6

Trace (plane section). The curve cut from the surface by a vertical plane such as $y = y_0$ (giving $z = f(x, y_0)$) or $x = x_0$ (giving $z = f(x_0, y)$). ADDED FOR CLARITY S002·7

Statements in the class material

> *"Partial derivative of the function f(x,y) at the point (x_0, y_0) can be visualized as the slope of the tangent line along the direction of the independent variable with respect to which we would differentiate the function."* S002·6

Key points

1. A surface has infinitely many tangents at a given point, one for every direction. ESA ★★ T002 00:05:24 · T002 00:05:35
2. z_x at (x_0, y_0) is the slope of the tangent to the surface at that point which is parallel to the x-axis. ESA ★★ T002 00:06:10 · T002 00:06:27
3. z_y at (x_0, y_0) is the slope of the tangent parallel to the y-axis. ESA ★★ T002 00:07:10 · T002 00:07:54
4. To find $\partial f/\partial x$ at (x_0, y_0): put $y = y_0$, which gives a curve $f(x, y_0)$ in one variable, then differentiate it at $x = x_0$. ESA ★★ S002·7
5. The cutting plane $y = y_0$ slices the surface in the curve whose slope is f_x; the plane $x = x_0$ gives the curve whose slope is f_y. ESA ★★ S002·7 · V002 00:07:45

Why?

Why are there infinitely many tangents at one point of a surface?
From the point you can move in infinitely many directions in the xy-plane. Each direction traces a different curve on the surface, and each curve has its own tangent at P.
T002 00:04:36 · T002 00:05:24

Why does fixing $y = y_0$ turn $\partial f/\partial x$ into an ordinary derivative?
Once $y = y_0$, $f(x, y_0)$ depends on x alone. It is a plane curve in the plane $y = y_0$, and its ordinary derivative at x_0 is the slope of its tangent.
S002·7

Exam traps

Exam trap. "Parallel to the x-axis" describes the tangent's direction seen from above (it lies in the plane $y = y_0$). It does not mean the tangent is horizontal: its slope is $f_x(x_0, y_0)$, which is usually not zero. ADDED FOR CLARITY

Exam trap. Hold the other variable constant before differentiating. Do not substitute $x = x_0$ before differentiating with respect to x. ADDED FOR CLARITY

10 Slope in any other direction: the gradient

Definitions

Gradient. The vector $\nabla f = f_x\,\mathbf{i} + f_y\,\mathbf{j}$: the resultant of the two slope components along the x- and y-axes. S002·8 · T002 00:09:23

Directional derivative. The slope of the surface in the direction of a unit vector $\hat{u}$: $D_{\hat{u}} f = \nabla f \cdot \hat{u}$. It is studied in vector calculus. ADDED FOR CLARITY T002 00:09:54

Statements in the class material

> *"Slope of tangent along any arbitrary direction (Gradient of a function). What about other directions than parallel to x and y?"* S002·8

Key points

1. f_x and f_y only give slopes in two directions; most tangents are parallel to neither axis. ESA ★ S002·8 · T002 00:08:22
2. Any line in the plane can be resolved into components along the x- and y-axes; vector addition of the components gives the resultant. ESA ★ T002 00:09:06 · T002 00:09:23
3. The resultant of $f_x\,\mathbf{i}$ and $f_y\,\mathbf{j}$ is the gradient ∇f, studied in vector calculus. ESA ★ S002·8 · T002 00:09:44
4. The slope in a unit direction $\hat{u} = (a, b)$ is $af_x + bf_y$. It is greatest, equal to $|\nabla f|$, in the direction of ∇f. ESA ★ ADDED FOR CLARITY S002·8

Why?

Why can't f_x or f_y alone give the slope along a diagonal direction?
Each is the slope along one axis only. A diagonal direction has both an x- and a y-component, so both slopes contribute. Combining them gives the gradient, and its dot product with the unit direction gives the slope.
S002·8 · T002 00:09:06

11 Second-order partials: concavity and mixed partials

Definitions

Concave up in the x direction. The trace in the x direction bends upward; the x-slope increases as x increases. Indicated by $f_{xx} > 0$. S002·9

Mixed partial f_{xy}. Tells how the rate of change of $f(x, y)$ in the x direction is changing as we move in the y direction. S002·9

Statements in the class material

> *"If $f_{xx} > 0$ then f(x,y) is concave up in the x direction. If $f_{yy} > 0$ then f(x,y) is concave up in the y direction. Mixed partials tell us how a partial in one variable is changing in the direction of the other. f_{xy} tells us how the rate of change of f(x,y) in the x direction is changing as we move in the y direction."* S002·9

Key points

1. As in one variable (concave up or down from d^2y/dx^2), the second-order partials describe how the surface bends. ESA ★★ T002 00:11:03 · T002 00:11:19
2. $f_{xx} > 0$ means concave up in the x direction; $f_{yy} > 0$ means concave up in the y direction. ESA ★★ S002·9 · T002 00:11:38
3. $f_{xx} < 0$ (or $f_{yy} < 0$) means concave down in that direction. One surface can be up in x and down in y, as in a saddle. ESA ★★ added for clarity S002·9
4. f_{xy} is the change in f_x along the y direction; f_{yx} is the change in f_y along the x direction. ESA ★★ S002·9 · T002 00:11:45 · T002 00:12:04
5. The class demo, "Second-Order Partial Derivatives", lets you set the signs of f_{xx}, f_{yy} and f_{xy} and watch the red tangent line move in 3D and contour views. ESA ★ S002·10 · V002 00:12:40 · T002 00:12:40

Why?

Why does $f_{xx} > 0$ mean concave up in x?
f_{xx} is the rate of change of the x-slope as x increases. If it is positive, the tangent keeps tilting upward as you move along x, so the trace bends upward.
S002·9 · T002 00:11:19

Why do the 4 second-order partials usually give only 3 different values?
For the smooth functions used in this unit the mixed partials are equal, $f_{xy} = f_{yx}$ (Clairaut), leaving only f_{xx}, f_{yy} and f_{xy}.
added for clarity T002 00:00:24

12 Problems on partial derivatives

Definitions

Clairaut's theorem. If f_{xy} and f_{yx} are continuous near a point, they are equal there. The class plans to verify it in the next session. TRANSCRIPT ONLY ADDED FOR CLARITY T002 00:14:59

Key points

1. Evaluate at the point only after all the differentiations are done. ESA ★★ ADDED FOR CLARITY CW U1 P1 · ESA Dec-23 1a
2. To verify $f_{xy} = f_{yx}$, work out each side separately and compare them. ESA ★★ CW U1 P2

Exam traps

Exam trap. In these problems $\log$ is the natural logarithm: $\frac{d}{dy}\log y = \frac{1}{y}$. ADDED FOR CLARITY

Exam trap. For x^y: with respect to x it is a power, $y\,x^{y-1}$; with respect to y it is an exponential, $x^y \log x$. ADDED FOR CLARITY

M3

Classes 003–004 · Total Derivative, Chain Rule and Implicit Functions

13 Composite functions and the one-variable chain rule

Definitions

Composite function. A function of a function: $f(g(h(x)))$, or a function of several variables whose variables are themselves functions of other variables. S003·4 · T003a 00:00:59

Key points

1. For one variable you differentiate a composite function from the outside in: f first, then g, then h. ESA ★ T003a 00:01:19
2. For several variables the first question is always "differentiate with respect to what?", because there is more than one variable. ESA ★ T003a 00:01:45

14 The total derivative rule

Definitions

Total derivative. If $z = f(x, y)$ where x and y are functions of one independent variable t, the ordinary derivative dz/dt is the total derivative of z with respect to t. S003·5 · T003b 00:00:52

Tree diagram. A diagram with z at the root, the intermediate variables in the middle and the final variable(s) at the leaves. Each branch is labelled with a derivative; multiply along each path and add over the paths. S003·5 · S003·14 · T003a 00:19:22 · T003a 00:25:02

Statements in the class material

> *"If $z = f(x, y)$ where x and y are functions of one independent variable 't' then we can find the derivative of z with respect t as* $\dfrac{dz}{dt} = \dfrac{\partial z}{\partial x}\dfrac{dx}{dt} + \dfrac{\partial z}{\partial y}\dfrac{dy}{dt}$*."* S003·5

Key points

1. You can always substitute $x(t)$ and $y(t)$ into z and differentiate directly, but this is often messy. The total derivative rule avoids substituting first. ESA ★★ T003a 00:02:39 · T003a 00:03:13
2. Both methods must give the same answer; substituting first is a good check on a short example. ESA ★★ (added for clarity) T003a 00:04:35
3. After using the rule, write the answer in terms of t alone (substitute $x(t)$, $y(t)$), or evaluate at the given t. ESA ★★ S003·19 · T003b 00:02:03
4. At a given t, a term vanishes if its dx/dt or dy/dt is 0 there (e.g. $dy/dt = \cos t = 0$ at $t = \pi/2$). ESA ★★ T003b 00:02:34

Why?

Why is it dz/dt (ordinary) and not $\partial z/\partial t$?
After substituting $x(t)$ and $y(t)$, z depends on the single variable t, so its derivative with respect to t is an ordinary derivative. The partial signs belong only to z with respect to x and y, where the other variable is held fixed.
S003·5 · T003a 00:02:23

Exam traps

Exam trap. Slide 003·6 has wrong numbers. For $z = x^2y^3$, $x = t^2$, $y = 2t^3$: $z_x = 2xy^3 = 16t^{11}$ (not $2t^{11}$), $z_y = 3x^2y^2 = 12t^{10}$ (not $9t^{10}$), and $dz/dt = 16t^{11}(2t) + 12t^{10}(6t^2) = 104t^{12}$ (not $60t^{12}$). Check: $z = 8t^{13}$. (added for clarity) S003·6

15 The total differential: approximation and sensitivity

Definitions

Total differential. For $z = f(x, y)$: $dz = f_x\,dx + f_y\,dy$. For $z = f(x_1, \ldots, x_n)$: $dz = f_{x_1}dx_1 + \cdots + f_{x_n}dx_n$. The total derivative is also called the differential. S004·3 · S004·4 · T004a 00:02:04

Statements in the class material

> *"The total derivative is also called the differential. The differential is used in integration by substitution and in approximation. It is also useful to calculate the Errors in approximation."* S004·4

Key points

1. The total differential estimates how much f changes from a known point to a nearby one. ESA ★★ S003·8
2. Sensitivity: in $dV = 40\pi\,dr + 4\pi\,dh$ the coefficient of dr is ten times that of dh, so the tank's volume is more sensitive to the radius than to the height. ESA ★★ S003·11 · S003·12 · T003a 00:18:30
3. Sometimes you need not finish the problem: the coefficients of the differential already tell you which variable contributes most. ESA ★ T003a 00:19:05
4. Integration by substitution uses the differential: with $t = x^2$, $dt = 2x\,dx$, so $\int e^{x^2} x\,dx = \frac{1}{2}\int e^t\,dt$. ESA ★ (transcript only) T004a 00:07:57 · T004a 00:08:29
5. Errors and approximations, another application of the differential, come at the end of the unit. ESA ★ (transcript only) T004a 00:08:47

Why?

Why does the tank's volume depend more on the radius?
$V = \pi r^2 h$ has $V_r = 2\pi r h = 40\pi$ and $V_h = \pi r^2 = 4\pi$ at $r = 2$, $h = 10$. So $dV = 4\pi(10\,dr + dh)$: a change in r is multiplied by ten times as much as the same change in h.
S003·11 · T003a 00:18:45

Exam traps

Exam trap. Use signed changes: from -3 to -3.03, $dy = -0.03$ (slide 003·8 writes 0.03 but correctly uses -0.03). S003·8

16 Rates of change problems

Definitions

Rate of change. The derivative of a quantity with respect to time t; positive means increasing, negative means decreasing. T003a 00:12:06

Key points

1. If $dz/dt > 0$ the quantity is increasing at that instant; if $dz/dt < 0$ it is decreasing. ESA ★★ T003a 00:12:06
2. When an unknown rate such as dy/dt is needed, find it from the constraint by implicit differentiation, e.g. $x^2 + y^2 = 5 \Rightarrow dy/dt = -\frac{x}{y}\,dx/dt$. ESA ★★ S003·7 · T003a 00:09:29
3. A decreasing side has a negative rate: "decreasing at 2 cm/s" means $dy/dt = -2$. ESA ★★ S003·10

Why?

Why does the angle have to decrease in the triangle problem?
$A = \frac{1}{2}xy\sin\theta$. One side grows (+3) and the other shrinks (−2), but at the given instant their combined effect $\frac{1}{2}\sin\theta(3y - 2x) = \frac{1}{2}\cdot\frac{1}{2}(90 - 40) > 0$ would increase the area. To keep A constant, θ must decrease.

ADDED FOR CLARITY S003·9 · S003·10

Exam traps

Exam trap. Slide 003·10 prints $d\theta/dt = 0.04831$. The value is $-25/(300\sqrt{3}) = -\sqrt{3}/36 \approx -0.0481$ rad/s: negative, and 481, not 4831. ADDED FOR CLARITY S003·10 · T003a 00:16:55

Exam trap. $x^2/16 - y^2/9 = z$ is a hyperbolic paraboloid (a saddle), not a cup-shaped paraboloid. The calculation does not change. ADDED FOR CLARITY S003·7 · T003a 00:06:27

17 The chain rule for composite functions

Definitions

Chain rule (several variables). If $z = f(x, y)$ is a bivariate function and x, y are functions of two or more independent variables u, v, then $\dfrac{\partial z}{\partial u} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial u} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial u}$ and $\dfrac{\partial z}{\partial v} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial v} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial v}$. S003·13 · T003a 00:21:07

The chain of a problem. The list of dependencies, e.g. "u depends on x, y; x, y depend on r, t; so u is a composite function of r, t". Knowing it fixes which chain-rule formula to write. T004b 00:02:31

Statements in the class material

> *"We can define chain rule for a function of two or more independent variable: If $z = f(x, y)$ is a bivariate function and x, y are functions of two or more independent variables say u, v then we can find the partial derivative of z with respect to u and v."* S003·13

Key points

1. When x and y depend on two variables u, v, z cannot be differentiated "totally": only z_u and z_v exist. ESA ★★ T003a 00:20:40
2. Write the tree diagram every time a function has more than one independent variable, so that no partial derivative is missed. ESA ★★★ T003a 00:25:02
3. It is essential to know the chain given in the problem: what depends on what. The formula follows from it. ESA ★★★ T004b 00:02:31
4. If f is not given ($u = f(x, y)$), keep f_x, f_y (or f_r, f_θ) as symbols in the answer. ESA ★★★ S003·17 · S003·25 · T003a 00:26:28
5. For $f(2x - 3y,\ 3y - 4z,\ 4z - 2x)$, name the arguments r, s, t and compute $r_x, s_x, t_x, \ldots$; most are constants or 0. ESA ★★★ S003·15 · T003a 00:22:33
6. When u is a composite function of r, θ (or s, t), eliminate x and y from the final answer: you must know why you substitute. ESA ★★★ T004b 00:07:52 · T004b 00:10:28
7. Functions of $x^2 + y^2$ alone (e.g. $\sin(x^2 + y^2)$, $e^{x^2+y^2}$) have $\partial u/\partial\theta = 0$ in polar coordinates, since $x^2 + y^2 = r^2$ does not involve θ. ESA ★★ (added for clarity) S004·10 · S004·13 · S004·14
8. $\tan^{-1}(y/x)$ is exactly θ in polar coordinates, so $\partial u/\partial\theta = 1$. ESA ★★ (added for clarity) S004·15
9. The chain rule extends to any number of variables: z of x, y, each of x, y of three variables, and so on. ESA ★ T003a 00:21:40

Why?

Why does $u = f(x - y,\ y - z,\ z - x)$ give $u_x + u_y + u_z = 0$?
Each argument is a difference whose derivatives sum to zero: for $r = x - y$, $r_x + r_y + r_z = 1 - 1 + 0 = 0$. Adding the three chain-rule expressions gives $f_r(0) + f_s(0) + f_t(0)$. In words: moving x, y, z by the same amount changes none of the arguments.
(added for clarity) ESA Jan-24 s2 1d

Why do the chain-rule answers keep f_r and f_θ?
The outer function f is unknown, so its partial derivatives cannot be computed; only the inner derivatives ($r_s = t$, $\theta_s = s/\sqrt{s^2+t^2}$, ...) can.

S003·17 · T003a 00:26:28

Exam traps

Exam trap. Slide 003·16 shows $\dfrac{\partial v}{\partial y} = -3\dfrac{\partial v}{\partial r} + \dfrac{\partial v}{\partial s}$; it should be $-3\dfrac{\partial v}{\partial r} + 3\dfrac{\partial v}{\partial s}$ because $s = 3y - 4z$ gives $s_y = 3$. With the 3 the proof works. (added for clarity) S003·16

Exam trap. Do not stop with x and y still in an answer that should be in r, θ or s, t. T004b 00:07:52

Exam trap. Write $\dfrac{\partial z}{\partial u}$, not $\dfrac{dz}{du}$: z depends on both u and v. (added for clarity) S003·13

18 Implicit functions of two variables

Definitions

Implicit function. A function given by a relation $f(x,y)=c$ (for example $x^3y+3xy^2=3$) rather than by a formula $y=g(x)$. S004·3 · T004a 00:00:18

Level surface (level curve). The set where $z=f(x,y)=c$ (or $f(x_1,\dots,x_n)=c$). Examples: equipotential surfaces, isotherms. The cylinders $x^2+y^2=c$ and the spheres $x^2+y^2+z^2=c$ are layers, one for each c. S004·4 · T004a 00:02:35 · T004a 00:03:29

Statements in the class material

> *"If z represents a level surface, i.e. if $z=f(x,y)=c$, then $dz=0$. The total derivative is also called the differential."* S004·4

> *"Setting $\dfrac{dz}{dx}=0$ leads to a simple computation formula for implicit differentiation: $f_x\cdot 1+f_y\dfrac{dy}{dx}=0$."* S004·5

Key points

1. Differentiating an implicit relation directly needs the product rule on every term, and the right-hand constant gives 0. ESA ★★ S004·3 · T004a 00:00:48
2. The formula $dy/dx=-f_x/f_y$ gives the same answer and is far easier when f is complicated. ESA ★★ T004a 00:06:19 · T004a 00:06:36
3. Restriction to remember: the formula needs $z=f(x,y)$ to be a constant (a level curve). Move everything to one side first: $f(x,y)=c$ or $f(x,y)=0$. ESA ★★★ T004a 00:01:49 · T004a 00:02:04
4. There are two ways to solve "find dy/dx": differentiate both sides directly, or use $-f_x/f_y$. Both are accepted. ESA ★★ S003·27 · T003b 00:10:33
5. The slope dy/dx at a point gives the tangent line $y-y_0=m(x-x_0)$. ESA ★★ S004·7 · T004a 00:12:01
6. For $(\cos x)^y=(\sin y)^x$ the result needs $\cos x>0$ and $\sin y>0$, because of the logarithms. ESA ★ T004a 00:18:12
7. Implicit and composite together: if $u=\tan(x^2+y^2)$ and $x^2-y^2=2$, use the relation to eliminate y (or find dy/dx) before differentiating. ESA ★★ CW U1 P16 · CW U1 P18

Why?

Why is $dy/dx=-f_x/f_y$?
Along the curve $f(x,y)=c$ the value of f does not change, so $df=f_x\,dx+f_y\,dy=0$. Divide by dx: $f_x+f_y\dfrac{dy}{dx}=0$. The branch diagram (Figure G18.1) shows the same sum.
S004·4 · S004·5 · T004a 00:07:39

Why is the level-curve slope $-f_x/f_y$ and not $-f_y/f_x$ (as printed on slide 004·4)?
From $f_x + f_y\,y' = 0$ you get $y' = -f_x/f_y$. Test it on $x^2 + y^2 = 5$ at $(2,1)$: $-f_x/f_y = -2x/(2y) = -2$, which matches differentiating directly; $-f_y/f_x$ would give $-1/2$.

ADDED FOR CLARITY S004·4 · S004·5

Exam traps

Exam trap. Slide 004·4 prints $dy/dx = -f_y(x,y)/f_x(x,y)$. The correct formula is $dy/dx = -f_x/f_y$. ADDED FOR CLARITY S004·4

Exam trap. In $(\cos x)^y$, with respect to x it is a power ($y(\cos x)^{y-1}(-\sin x)$); with respect to y it is an exponential ($(\cos x)^y \ln\cos x$). S004·8 · T004a 00:16:02

19 Implicit functions of three variables

Definitions

Implicit function of three variables. $f(x, y, z) = 0$ with z (or y) regarded as a function of the remaining variables; the other variables are intermediate variables. S004·6 · T004a 00:09:05

Statements in the class material

> *"Let $f(x, y, z) = 0$ be the equation of an implicit function of three variables x, y, z. Suppose y and z are functions of x, then f is a function of one independent variable x and y, z are intermediate variables."* S004·6

Key points

1. For $f(x, y, z) = 0$, x and y are independent, so dz alone has no meaning; find the partials z_x, z_y. ESA ★★ T004a 00:09:38
2. For $\partial z/\partial x$ treat y as constant; for $\partial y/\partial x$ treat z as constant. ESA ★★ S004·6 · S004·9 · T004a 00:19:00
3. The formula avoids separating z from a messy equation such as $x^3y^2 + 3xyz + \sin(e^z) = c$. ESA ★ T004a 00:10:55
4. Check that the given point satisfies $f = 0$ before substituting. ESA ★★ (added for clarity) S004·9 · T004a 00:18:39

Exam traps

Exam trap. The lecture works problem 3 at $(1, 0, 2)$ and gets 0 for both derivatives, but $(1, 0, 2)$ is not on $z^3 + xy - y^2z = 6$ (it gives 8). Use the slide's point $(0, 1, 2)$: $\partial y/\partial x = 1/4$, $\partial z/\partial x = -1/11$. (added for clarity) S004·9 · T004a 00:20:47

Exam trap. Do not differentiate $-f_x/f_z$ as if z were a constant when you need second derivatives: z is still a function of x and y (W19.3). (added for clarity) ESA Jan-24 s3 1d

M4

Class 005 · Homogeneous Functions and Euler's Theorem

20 Homogeneous functions

Definitions

Homogeneous function (two variables). A function $u = f(x, y)$ is said to be a homogeneous function of degree n if it can be expressed in the form $x^n g\left(\frac{y}{x}\right)$ or $y^n g\left(\frac{x}{y}\right)$, g being an arbitrary function. S005·3 · T005 00:02:45

Homogeneous function (three variables). $u = f(x, y, z)$ is homogeneous of degree n if it can be expressed as $x^n g\left(\frac{y}{x}, \frac{z}{x}\right)$, or $y^n g\left(\frac{x}{y}, \frac{z}{y}\right)$, or $z^n g\left(\frac{x}{z}, \frac{y}{z}\right)$. S005·4 · T005 00:04:38

Scaling test. f is homogeneous of degree n exactly when $f(tx, ty) = t^n f(x, y)$ for all $t > 0$: scaling every variable by t scales the function by t^n. (added for clarity) S005·5

Degree. The power n in $x^n g(y/x)$; for a polynomial whose terms all have the same total degree, that common degree. T005 00:01:31

Key points

1. For a polynomial, check that every term has the same total degree; that is the degree ($x^3 + 3x^2y + 3xy^2 + y^3$: degree 3). ESA ★★ T005 00:01:15 · T005 00:01:31
2. To test any function: take out x^n (or y^n) and check that what remains depends only on y/x (and z/x). ESA ★★ S005·3 · T005 00:05:42
3. Either form, $x^n g(y/x)$ or $y^n h(x/y)$, may be used — whichever is convenient — and both give the same degree. ESA ★★ T005 00:04:01
4. Inverse trigonometric or exponential functions of a homogeneous expression are usually not homogeneous; first find an f with $f(u)$ homogeneous (topic 4). ESA ★★★ T005 00:01:47 · T005 00:02:04
5. Homogeneous functions appeared earlier in homogeneous differential equations, solved by $y = vx$. ESA ★ (transcript only) T005 00:00:22

Exam traps

Exam trap. The lecture says the converse holds: "a function of degree 0 should be a constant function". It does not: y/x, $\sin^{-1}(y/x)$ and $\frac{x-y}{x+y}$ all have degree 0 and are not constant.
(added for clarity) T005 00:07:11

Exam trap. "Homogeneous" does not mean "smooth everywhere" (as said at T005 00:11:06); it is a scaling property. $(\sqrt{x}+\sqrt{y})/(x+y)$ is homogeneous but not defined at the origin.

ADDED FOR CLARITY T005 00:11:06

21 Euler's theorem

Statements in the class material

> *"Statement 1: If $u = f(x, y)$ is a homogeneous function of degree n then $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = nu$."* S005·6

> *"Statement 2: If $u = f(x, y, z)$ is a homogeneous function of degree n then $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z} = nu$."* S005·6

Key points

1. You do not need to differentiate: check that u is homogeneous, find n, and write $xu_x + yu_y = nu$. ESA ★★★ T005 00:09:17 · T005 00:17:32
2. The lecturer calls Euler's theorem "the most important one" of the session. ESA ★★★ T005 00:08:57
3. Check: $u = x^2 + y^2$ gives $x(2x) + y(2y) = 2u$, degree 2. ESA ★★ T005 00:10:31
4. Meaning: the change along x scaled by x, plus the change along y scaled by y, equals the function scaled by its degree. ESA ★ T005 00:09:46 · T005 00:11:24

Why?

Why is $xu_x + yu_y = nu$? (A short proof.)
Differentiate $u(tx, ty) = t^n u(x, y)$ with respect to t using the chain rule: $x\,u_x(tx, ty) + y\,u_y(tx, ty) = nt^{n-1}u(x, y)$. Put $t = 1$.

ADDED FOR CLARITY S005·6

22 Second and higher order: the corollary

Statements in the class material

> *"Corollary 1: If $u = f(x, y)$ is a homogeneous function of degree n then $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = n(n-1)u$."* S005·6

Key points

1. The mixed partials u_{xy} and u_{yx} are equal here (continuous derivatives), which is why the middle term is $2xy\,u_{xy}$. ESA ★★ T005 00:12:04
2. For $u = u_1 + u_2$ with different degrees n_1, n_2, apply the results to each part: $x^2 u_{xx} + \cdots = n_1(n_1 - 1)u_1 + n_2(n_2 - 1)u_2$. ESA ★★★ ADDED FOR CLARITY ESA Jan-24 s3 1c · CW U1 P22
3. Extend to three or more variables as an exercise; the lecture leaves it open and points to a research paper on higher-order extensions. ESA ★ S005·6 · T005 00:14:59

Why?

Why is the second-order right side $n(n-1)u$?
u_x and u_y are homogeneous of degree $n - 1$, so Euler's theorem applied to them gives $xu_{xx} + yu_{xy} = (n-1)u_x$ and $xu_{yx} + yu_{yy} = (n-1)u_y$. Multiply by x and y, add, and use $xu_x + yu_y = nu$.

ADDED FOR CLARITY S005·6

23 Extension: when f(u) is homogeneous

Statements in the class material

> *"If $f(u)$ be a homogeneous function of degree n in x and y, then* $x\dfrac{\partial u}{\partial x}+y\dfrac{\partial u}{\partial y}=n\dfrac{f(u)}{f'(u)}$ *and* $x^2\dfrac{\partial^2 u}{\partial x^2}+2xy\dfrac{\partial^2 u}{\partial x\partial y}+y^2\dfrac{\partial^2 u}{\partial y^2}=g(u)[g'(u)-1]$ *where* $g(u)=n\dfrac{f(u)}{f'(u)}$*."* S005·7

Key points

1. Spot the extension case: $u=\sin^{-1}(\dots)$, $\tan^{-1}(\dots)$, $\log(\dots)$ or $e^{(\cdots)}$ of a homogeneous expression; then f is $\sin$, $\tan$, $\exp$ or $\log$ respectively. ESA ★★★ (added for clarity) S005·7 · ESA May-24 1a
2. The degree n belongs to $f(u)$, not to u. ESA ★★★ S005·7

Why?

Where does $n\,f(u)/f'(u)$ come from?
Apply Euler to $F=f(u)$: $xF_x+yF_y=nF$. By the chain rule $F_x=f'(u)u_x$, $F_y=f'(u)u_y$, so $f'(u)(xu_x+yu_y)=nf(u)$.
(added for clarity) S005·7

Exam traps

Exam trap. For $u=\tan^{-1}\frac{x^4+y^4}{xy}$ the degree is that of $\tan u=\frac{x^4+y^4}{xy}$, i.e. $4-2=2$, not of u.
(added for clarity) ESA May-24 1a

24 Applications of homogeneous functions

Definitions

Extensive / intensive variable. Extensive variables (mass, volume, energy) double when the system is doubled: homogeneous of degree 1. Intensive variables (pressure, temperature, density) do not change: degree 0. (added for clarity) S005·5 · T005 00:16:26

Statements in the class material

> *"A constant function is homogeneous of degree 0. Linear functions are homogeneous of degree 1. The degree of the homogeneous function can be positive, negative or zero."* S005·5

Key points

1. A constant function is homogeneous of degree 0; linear functions are homogeneous of degree 1. ESA ★ S005·5 · T005 00:06:52
2. Euler's theorem also describes the scaling of the function: $n\,u$. ESA ★ S005·5 · T005 00:15:57
3. Applications: production functions, marginal productivities and elasticity of demand in economics; extensive (degree 1) and intensive (degree 0) variables in thermodynamics. ESA ★ S005·5 · T005 00:16:09
4. Any function of ratios only, such as $f(x/y,\, y/z,\, z/x)$, has degree 0, so $xu_x + yu_y + zu_z = 0$ (ESA Jan 2024 set 1). ESA ★★★ (added for clarity) ESA Jan-24 s1 1d

Exam traps

Exam trap. The lecture swaps the examples: mass is extensive (degree 1); pressure and temperature are intensive (degree 0). (added for clarity) T005 00:16:46

M5

Class 006 · Problems on Euler's Theorem

25 Testing homogeneity: the scaling test

Definitions

Homogeneous function (scaling form). $u(x, y)$ is homogeneous of degree n if $u(\lambda x, \lambda y) = \lambda^n u(x, y)$. Here n can be positive, negative or even zero. S006·5 · T006b 00:00:52 · T006b 00:01:11

Not homogeneous. u is not homogeneous when $u(\lambda x, \lambda y) \neq \lambda^n u(x, y)$ for every n: no single power of λ can be taken out. S006·8 · T006b 00:04:59

Key points

1. Write $u(\lambda x, \lambda y)$, take the highest power of λ out of the numerator and the denominator, and cancel. ESA ★★ S006·5 · T006b 00:02:17
2. A λ inside a ratio cancels: $\sin^{-1}\frac{\lambda y}{\lambda x} = \sin^{-1}\frac{y}{x}$, so functions of y/x add degree 0. ESA ★★ S006·6 · T006b 00:03:34
3. $e^{\lambda x}$ cannot be written as $\lambda^n e^x$, so exponentials of x or y alone break homogeneity (slide 006·8). ESA ★★ S006·8 · T006b 00:04:40
4. Simplify first when you can: slide 006·5 is $\frac{(x^2+y^2)^2}{(x^2+y^2)^2} = 1$ and slide 006·7 is $\frac{xy(x+y)}{(x+y)^3} = \frac{xy}{(x+y)^2}$; both have degree 0. ESA ★ ADDED FOR CLARITY S006·5 · S006·7

Why?

Why does slide 006·8 fail the test?

$u(\lambda x, \lambda y) = \dfrac{e^{\lambda x} - e^{\lambda y}}{\lambda(x - y)}$. The denominator gives λ^{-1}, but $e^{\lambda x} - e^{\lambda y}$ is not a power of λ times $e^x - e^y$. Figure G25.1 shows the ratio is not a straight line on log-log axes.

ADDED FOR CLARITY S006·8

26 Verifying Euler's theorem directly

Statements in the class material

> *"Euler's theorem: if $u = f(x, y)$ is a homogeneous function of degree n, then $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = nu$."*
>
> S006·10 · T006a 00:00:00

Key points

1. Verification is a check of the theorem, not a method: the lecture spends four minutes on slide 006·3; applying the theorem takes one line. ESA ★ T006a 00:04:57 · T006a 00:08:41
2. In slide 006·3, group the terms with the common denominator $(\sqrt{x} + \sqrt{y})^2$ and cancel one factor of $\sqrt{x} + \sqrt{y}$. ESA ★ T006a 00:02:59 · T006a 00:03:17

Exam traps

Exam trap. Slide 006·3 says "verify for $\cos^{-1}(\dots)$" but verifies for $Z = \cos u$. For u itself, $xu_x + yu_y = \frac{1}{2}\frac{\cos u}{-\sin u} = -\frac{1}{2}\cot u$, not $\frac{1}{2}u$. (ADDED FOR CLARITY) S006·3

27 When u itself is homogeneous: first and second order

Statements in the class material

> *"By Euler's theorem for a homogeneous function of degree n: $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$."* S006·11

Key points

1. First prove that u is homogeneous and find n; then write both answers without differentiating. ESA ★★★ T006b 00:06:11 · T006b 00:06:45
2. Give the answer in terms of x and y too: $-u = -\frac{x-y}{x^2+y^2}$ and $2u = \frac{2(x-y)}{x^2+y^2}$. ESA ★★ S006·10 · S006·11
3. For a sum of parts with different degrees, apply the corollary part by part: $n_1(n_1-1)u_1 + n_2(n_2-1)u_2$. ESA ★★★ (added for clarity) ESA Jan-24 s3 1c · CW U1 P22

Exam traps

Exam trap. $n = -1$ gives $n(n-1) = (-1)(-2) = +2$, not -2: the second-order answer on slide 006·11 is $+2u$. (added for clarity) S006·11

28 When f(u) is homogeneous: the extension

Definitions

Extension function g(u). $g(u) = n\dfrac{f(u)}{f'(u)}$, where $f(u)$ is homogeneous of degree n in x and y.

S006·13 · S006·15 · S006·17

Statements in the class material

> *"For $f(u)$ a homogeneous function of degree n: $xu_x + yu_y = n\left(\dfrac{f(u)}{f'(u)}\right) = g(u)$, and $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = g(u)\,[g'(u) - 1]$."* S006·13 · S006·15 · S006·17

Key points

1. When u is $\sin^{-1}$, $\tan^{-1}$, $\log$ or e of something, apply the inverse function to both sides: that is $f(u)$. ESA ★★★ T006b 00:08:20 · T006b 00:09:55 · T006b 00:11:40
2. The degree belongs to $f(u)$, found by the scaling test on $f(u)$ (λ^4 on slide 006·12, λ on 006·14, $\lambda^{1/2}$ on 006·16). ESA ★★★ S006·12 · S006·14 · S006·16
3. "Whenever you have a complicated function, you can always use Euler's theorem if you feel that the function is homogeneous" — the lecture's closing advice for part I. ESA ★★ T006a 00:08:53
4. Three variables work the same way: on slide 006·4, $\sin u$ has degree -3, so $xu_x + yu_y + zu_z = -3\tan u$. ESA ★★ S006·4 · T006a 00:06:54

Why?

Why does the log case give -1 for the second order?
$f(u) = e^u$ gives $g = n\,e^u/e^u = n$, a constant. So $g' = 0$ and $g(g' - 1) = -n$; slide 006·14 has $n = 1$.

S006·15

Why can't Euler's theorem be applied to u directly on slides 006·12–17?
u is not homogeneous: for $u = \log\frac{x^2+y^2}{x+y}$, $u(\lambda x, \lambda y) = \log\lambda + u$, which is a shift, not a factor λ^n.
ADDED FOR CLARITY T006b 00:09:55

Exam traps

Exam trap. For $u = e^{(\cdots)}$ the function is $f(u) = \log u$, so $f' = 1/u$ and $f/f' = u\log u$: the answer is $4u\log u$, not $4\log u$. ADDED FOR CLARITY S006·13

Exam trap. In the second order use $g'(u)$, the derivative of g with respect to u, not $f'(u)$: for $g = \frac{1}{2}\tan u$, $g' = \frac{1}{2}\sec^2 u$. S006·17 · T006b 00:12:42

M6

Class 007 · Taylor's and Maclaurin's Expansion

29 Why Taylor's and Maclaurin's series?

Key points

1. Once a function is a polynomial, you can differentiate, integrate, find limits and study how it changes. ESA ★ T007 00:06:50
2. Applications (one variable): polynomials for complicated functions, definite integrals with no closed form, asymptotic behaviour, growth of functions, differential equations, the Lorentz factor, $E = mc^2$ at low speed, $\sin\theta \approx \theta$ for a pendulum, power-flow analysis, generating functions. ESA ★ S007·6 · S007·7 · T007 00:11:31 · T007 00:13:44
3. Applications (two variables): optimization (a linear or quadratic model), accurate distance measurement with the sine law, variance of complicated configurations, visualizing complicated functions. ESA ★ S007·12 · T007 00:21:22
4. A few terms are often enough: four terms of the sine series already follow $\sin x$ closely near 0, and ten terms overlap it (Figure G29.1). ESA ★ S007·8 · T007 00:15:24 · T007 00:16:10
5. Taylor's theorem is the basis of L'Hôpital's rule and the mean value theorems of Rolle, Lagrange and Cauchy; the series is a consequence of the theorem. ESA ★ TRANSCRIPT ONLY T007 00:03:33

Why?

Why does a calculator need series at all?
It can only add and multiply. A polynomial needs nothing else, so a function such as $e^{\tan x}$ is evaluated through polynomial approximations.
T007 00:01:02 · T007 00:01:35

Exam traps

Exam trap. Slide 007·3's limit $\lim \frac{\sin x - x}{x^4}$ does not exist; with x^3 the answer is $-\frac{1}{6}$ (W29.2).
ADDED FOR CLARITY S007·3

30 Taylor's and Maclaurin's series in one variable

Definitions

Taylor's series (one variable). If a function $f(x)$ has continuous derivatives of order $(n+1)$, then this function can be expressed as a polynomial at a point $x = a$ in the form $f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \cdots + \frac{(x-a)^n}{n!}f^{(n)}(a) + \ldots$ S007·4 · T007 00:04:19

Maclaurin's series (one variable). If the function is expanded about the origin we get Maclaurin's series: $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots + \frac{x^n}{n!}f^{(n)}(0) + \ldots$; it is a particular case of Taylor's series. S007·4 · T007 00:06:17

Key points

1. Keep only the first two terms of Taylor's series and you get $f(x) \approx f(a) + (x-a)f'(a)$: the tangent line, i.e. the derivative as a slope (Figure G30.1). ESA ★ T007 00:05:26 · T007 00:05:43
2. To get a standard series, differentiate repeatedly and put $x = 0$: every derivative of e^x is e^x, which is 1 at the origin. ESA ★ T007 00:09:10 · T007 00:09:44
3. Substitute into a known series rather than differentiating: $\sin(x^2) = x^2 - \frac{x^6}{3!} + \frac{x^{10}}{5!} - \ldots$. ESA ★★ T007 00:11:51
4. No Maclaurin series exists if f or a derivative is infinite at 0: $1/x$, $\log x$, $\sqrt{x}$. ESA ★ T007 00:10:18

Why?

When is Maclaurin's series valid?
f must have derivatives of every order at 0 (so none is infinite there), and the series must converge to f. e^x, $\sin x$, $\cos x$ converge for every x; $\log(1+x)$ and $(1+x)^n$ only for $|x| < 1$.

added for clarity T007 00:10:01 · S007·5

Exam traps

Exam trap. The lecture says $1 + 2 + 2^2 + \ldots$ "gives -1". It diverges: $\frac{1}{1-x} = 1 + x + x^2 + \ldots$ only for $|x| < 1$; $\frac{1}{1-2} = -1$ is outside that range. added for clarity T007 00:07:52

Exam trap. $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \ldots$ has plain denominators; slide 007·5 shows $3!$ and $5!$, which belong to $\sin x$. added for clarity S007·5

31 Taylor's expansion of a function of two variables

Definitions

Taylor polynomial of degree n. The terms of Taylor's series up to total degree n in $(x-a)$ and $(y-b)$; "expand up to second (third) degree terms" means stop there. Degree 1 is the tangent plane. ADDED FOR CLARITY S007·11 · T007 00:20:13

Statements in the class material

> *"Taylor's series of $f(x,y)$ about (a,b) is given by* $f(x,y) = f(a,b) + \frac{1}{1!}\{(x-a)f_x(a,b) + (y-b)f_y(a,b)\} + \frac{1}{2!}\{(x-a)^2 f_{xx}(a,b) + 2(x-a)(y-b)f_{xy}(a,b) + (y-b)^2 f_{yy}(a,b)\} + \ldots$" S007·9 · T007 00:18:45

Key points

1. Lay out a table: f, f_x, f_y, f_{xx}, f_{xy}, f_{yy} (and third order if asked), each in general and then its value at (a,b). Then substitute into E31.1. ESA ★★★ ADDED FOR CLARITY CW U1 P24 · ESA Jul-23 1b
2. There are four second-order partials, but $f_{xy} = f_{yx}$, so they combine into $2(x-a)(y-b)f_{xy}$. ESA ★★ T007 00:18:26
3. "In powers of $(x-1)$ and $(y-\frac{\pi}{4})$" means about the point $(1, \frac{\pi}{4})$. ESA ★★ ADDED FOR CLARITY ESA May-24 1b · CW U1 P27
4. Taylor's series gives approximate values: $\tan^{-1}\frac{0.9}{1.1}$ from the expansion of $\tan^{-1}\frac{y}{x}$ about $(1,1)$ with $h = 0.1$, $k = -0.1$. ESA ★★ CW U1 P26
5. The first-degree Taylor polynomial is the tangent plane; the second degree hugs the surface much better (slide 007·11, Figures G31.1–G31.2). ESA ★ S007·11 · T007 00:20:51

Why?

Why are the second-order coefficients 1, 2, 1?
Put $g(t) = f(a+th, b+tk)$ and use the one-variable Maclaurin series of g at $t=0$. By the chain rule $g'(0) = hf_x + kf_y$ and $g''(0) = h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}$ (the two equal mixed partials give the 2). Put $t = 1$.
ADDED FOR CLARITY T007 00:18:26

Exam traps

Exam trap. Evaluate every derivative at (a,b) before substituting; never leave $f_{xx}(x,y)$ in the answer. ADDED FOR CLARITY S007·9

Exam trap. Don't forget $\frac{1}{2!}$ and $\frac{1}{3!}$, and the middle coefficients **2** and **3**: the third-order bracket is $h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}$. ADDED FOR CLARITY ESA Dec-23 1b

Exam trap. Slide 007·11's polynomials ($1 + x + y$, $1 + x + xy + \frac{y^2}{2}$) are about $(1, 0)$, not the origin; about the origin they are $1 + x$ and $1 + x + xy$. ADDED FOR CLARITY S007·11 · T007 00:20:32

32 Maclaurin's expansion of a function of two variables

Definitions

Maclaurin's series (two variables). Taylor's series of $f(x,y)$ about the origin $(0,0)$. S007·10 · T007 00:19:01

Statements in the class material

> *"Maclaurin's series for a function of two variables is given by* $f(x,y) = f(0,0) + \frac{1}{1!}\{xf_x(0,0) + yf_y(0,0)\} + \frac{1}{2!}\{x^2 f_{xx}(0,0) + 2xyf_{xy}(0,0) + y^2 f_{yy}(0,0)\} + \ldots$*"* S007·10

Key points

1. Memory aid from the lecture: f_x goes with x, f_y with y, f_{xx} with x^2, f_{xy} with $2xy$, f_{yy} with y^2. ESA ★★ T007 00:19:38
2. Shortcut and check: if f depends on one combination, use the one-variable series: $\sin(x+2y) = (x+2y) - \frac{(x+2y)^3}{6} + \ldots$, $\log(1+x-y) = (x-y) - \frac{(x-y)^2}{2} + \frac{(x-y)^3}{3} - \ldots$ ESA ★★★ (added for clarity) CW U1 P25 · ESA Jun-24 1c · ESA Jan-24 s1 1b
3. None of f, f_x, f_y, f_{xx}, f_{xy}, f_{yy} may be infinite at $(0,0)$. ESA ★ T007 00:19:26

M7

Class 008 · Problems on Taylor's and Maclaurin's Series

33 Taylor expansion about a point (a, b)

Definitions

Expansion "in powers of (x − a) and (y − b)". Taylor's series of $f(x,y)$ about the point (a,b). S008·3 · T008 00:00:37

Up to n-th degree terms. Keep every term whose total power of $(x-a)$ and $(y-b)$ is at most n. ADDED FOR CLARITY T008 00:06:10

Statements in the class material

> "$f(x,y) \approx f(a,b) + (x-a)f_x(a,b) + (y-b)f_y(a,b) + \frac{1}{2!}\left[(x-a)^2 f_{xx}(a,b) + 2(x-a)(y-b)f_{xy}(a,b) + (y-b)^2 f_{yy}(a,b)\right]$" S008·14 · S008·19

Key points

1. Write the general series first, then find the derivatives, then substitute. That is the order the lecture follows. ESA ★★ T008 00:00:21 · T008 00:02:07
2. For $e^x \cos y$ at $(1, \frac{\pi}{4})$ every derivative is $\pm\frac{e}{\sqrt{2}}$; only the signs differ, so take $\frac{e}{\sqrt{2}}$ out as a common factor. ESA ★★★ T008 00:04:54 · T008 00:05:19
3. If the question asks for third degree, you need the third-order partials $f_{xxx}, f_{xxy}, f_{xyy}, f_{yyy}$ too. ESA ★★★ T008 00:06:10
4. Derivatives that vanish at the point drop whole terms: for $\sin(xy)$ at $(1, \frac{\pi}{2})$ both first partials are 0. ESA ★★ S008·15 · S008·16
5. Differentiate carefully with the product and chain rules; the slide's own f_{yy} for $xy^2 + y\cos(x-y)$ has a sign slip (see the exam traps). ESA ★★ ADDED FOR CLARITY S008·17 · S008·18

Exam traps

Exam trap. For $xy^2 + y\cos(x-y)$: $f_{yy} = 2x + 2\sin(x-y) - y\cos(x-y)$, so $f_{yy}(1,1) = 1$, not 3 as slide 008·18 prints. ADDED FOR CLARITY S008·17 · S008·18

Exam trap. Every value on slide 008·3 is $\pm\frac{e}{\sqrt{2}}$, not $\pm\frac{e}{2}$ as said at T008 00:03:47.

ADDED FOR CLARITY S008·3 · T008 00:03:47

34 Maclaurin expansion about the origin

Definitions

Up to three terms. Keep going until there are three non-zero terms: "make sure that you will get three non-zero terms". S008·5 · T008 00:07:38

Statements in the class material

> *"Taylor's series about* $(0,0)$ *is given by* $f(x,y) \approx f(0,0) + f_x(0,0)\,x + f_y(0,0)\,y + \frac{1}{2!}\left[f_{xx}(0,0)x^2 + 2f_{xy}(0,0)xy + f_{yy}(0,0)y^2\right] + \frac{1}{3!}\left[f_{xxx}(0,0)x^3 + 3f_{xxy}(0,0)x^2y + 3f_{xyy}(0,0)xy^2 + f_{yyy}(0,0)y^3\right] + \ldots$" S008·10

Key points

1. About $(0,0)$ the constant term is often 0 ($\log 1 = 0$, $\sin 0 = 0$), so the series starts at first degree. ESA ★★ T008 00:07:57
2. "Up to three terms" means three non-zero terms; the lecture still writes all second-order terms. ESA ★★ T008 00:07:38 · T008 00:10:55
3. Collect and simplify at the end: $2x + y - 2x^2 - 2xy - \frac{y^2}{2}$. ESA ★ T008 00:11:11
4. Check a two-variable answer with the one-variable series of the combination (E34.2, Figure G34.1). ESA ★★★ added for clarity S008·6 · S008·11

Why?

Why does $\log(2x + y + 1)$ give $(2x + y) - \frac{(2x+y)^2}{2} + \ldots$?
It is $\log(1 + t)$ with $t = 2x + y$, and $\log(1 + t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \ldots$; the two-variable Maclaurin series must agree with it.
added for clarity S008·6

Exam traps

Exam trap. For $e^x \log(1 + y)$: $f_{xy} = \frac{e^x}{1+y}$, so $f_{xy}(0,0) = 1$, not 0 as slide 008·12 lists.
added for clarity S008·12

Exam trap. "First six terms" of $e^x \log(1 + y)$ needs third order: $y + xy - \frac{y^2}{2} + \frac{x^2y}{2} - \frac{xy^2}{2} + \frac{y^3}{3}$. The slide stops at three. added for clarity S008·12 · S008·13

35 Approximate values from a Taylor polynomial

Definitions

Approximate value. The value of a truncated Taylor polynomial at a point near the expansion point. S008·7 · T008 00:11:47

Key points

1. Pick an expansion point near the target where f is easy: $(1,-1)$ for $(0.9,-1.2)$, $(1,1)$ for $(1.1,0.9)$. ESA ★★ T008 00:12:10 · CW U1 P26
2. "The more terms you take, the better the approximation": the first-order estimate of $\tan^{-1}(-1.08)$ is off by 0.012, the second-order one by 0.001. ESA ★★ T008 00:14:18
3. Substitute $h = x - a$ and $k = y - b$ with their signs: $x = 0.9$ gives $h = -0.1$, $y = -1.2$ gives $k = -0.2$. ESA ★★ T008 00:15:13 · T008 00:15:28

Why?

Why expand about (1, −1) to evaluate at (0.9, −1.2)?
At $(1,-1)$ every value is exact ($\tan^{-1}(-1) = -\frac{\pi}{4}$, simple derivatives), and $h = -0.1$, $k = -0.2$ are small, so the higher terms shrink quickly.
T008 00:12:10 · T008 00:12:48

Exam traps

Exam trap. With $x = 0.9$, $h = x - 1 = -0.1$ (negative). A sign slip here changes the answer by 0.1. (added for clarity) T008 00:15:13

M8

Class 009 · Maxima and Minima of a Function of Two Variables

36 Maxima, minima and saddle points

Definitions

Saddle point. A stationary point at which $f(a,b)$ is neither a maximum nor a minimum: the surface rises in some directions and falls in others. Test: $rt - s^2 < 0$. S009·5 · S009·7 · T009 00:11:06

Key points

1. One variable recap: solve $\frac{dy}{dx} = 0$ for critical points; $\frac{d^2y}{dx^2} < 0$ gives a maximum, > 0 a minimum; $= 0$ gives no conclusion. The tangent at a maximum or minimum is parallel to the x-axis (slope 0). ESA ★★ T009 00:00:50 · T009 00:01:40 · T009 00:02:57
2. In two variables the tangent plane at a maximum or minimum is horizontal (parallel to the xy-plane); slide 009·3 shows a horizontal tangent plane at a relative maximum (Figure G36.4). ESA ★ S009·3 · S009·4 · T009 00:01:10
3. A function can have several local maxima and several local minima (slides 009·3–4: "These are all local maxima / minima"). ESA ★ S009·3 · S009·4 · T009 00:00:16

Exam traps

Exam trap. In one variable $\frac{d^2y}{dx^2} = 0$ is not "a saddle point"; it is only inconclusive (x^4 has a minimum, x^3 an inflection). (added for clarity) T009 00:03:14

37 Definitions: relative maximum, minimum and extreme value

Definitions

Relative maximum. A function $f(x, y)$ is said to have a relative maximum at a point (a, b) if there exists a neighbourhood of the point (a, b), say $(a + h, b + k)$, h and k small, such that $f(a, b) > f(a + h, b + k)$. S009·6 · T009 00:03:46

Relative minimum. If $f(a, b) < f(a + h, b + k)$ then $f(x, y)$ is said to have a relative minimum at (a, b). S009·6 · T009 00:04:58

Extreme value. $f(a, b)$ is said to be an extreme value of f if it is either a maximum or a minimum. S009·6

Neighbourhood. The points very close to (a, b): $(a + h, b + k)$ with h, k small. "Relative" means largest or smallest only in such a neighbourhood, not over the whole plane. T009 00:04:12 · T009 00:04:40

Statements in the class material

> *"In other words if $\Delta = f(a + h, b + k) - f(a, b)$ is of the same sign for all small values of h, k and if this sign is negative then $f(a, b)$ is maximum & if this sign is positive then $f(a, b)$ is a minimum."* S009·6 · T009 00:05:23

Key points

1. "Relative" because only points in a neighbourhood of (a, b) are compared, not the whole plane. ESA ★ T009 00:04:40

38 Conditions and the working rule

Definitions

Stationary (critical) point. A point where $f_x = 0$ and $f_y = 0$: the tangent plane is horizontal, and the function may have an extreme value there. S009·8 · T009 00:02:47 · T009 00:06:19

Hessian. The determinant $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = rt - s^2$ of second partial derivatives. S009·7 · T009 00:07:22

Statements in the class material

> *"The necessary and sufficient condition for $f(x, y)$ to have a maximum value at (a, b) is $f_x(a, b) = 0$, $f_y(a, b) = 0$ and $rt - s^2 > 0$, where $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$. $rt - s^2$ is the determinant of the Hessian matrix $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix}$."* S009·7

> *"Note: If $rt - s^2 < 0$, $f(a, b)$ is not an extreme value, i.e. $f(x, y)$ is neither a maximum nor a minimum at the point (a, b) and the point is called a saddle point. If $rt - s^2 = 0$, then the test provides no information about the nature of the point."* S009·7

Key points

1. Working rule step 1: find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$, equate each to zero, solve simultaneously; let $(a, b), (c, d), \ldots$ be the pairs of values. ESA ★★★ S009·8 · T009 00:11:32
2. Step 2: calculate $r = \frac{\partial^2 f}{\partial x^2}$, $s = \frac{\partial^2 f}{\partial x \partial y}$, $t = \frac{\partial^2 f}{\partial y^2}$ for each pair of values. ESA ★★★ S009·8
3. Steps 3–6: $rt - s^2 > 0$ and $r < 0$: maximum; $rt - s^2 > 0$ and $r > 0$: minimum; $rt - s^2 < 0$: not an extreme value, (a, b) is a saddle point; $rt - s^2 = 0$: the case is doubtful and needs further investigation. ESA ★★★ S009·8 · S009·9 · T009 00:12:33
4. Two important conditions to remember (lecture): both first partials zero, and $rt - s^2$ from the second partials. ESA ★★★ (transcript only) T009 00:08:50
5. Unlike one variable, the sign of r alone is not enough: first check $rt - s^2 > 0$, then look at r. ESA ★★★ T009 00:07:51 · T009 00:09:43
6. The value of the function at a maximum or minimum point is the maximum or minimum value: always substitute the point back into f. ESA ★★ (added for clarity) T009 00:04:51
7. Each stationary point is tested separately: a function can have maxima, minima and saddle points together (class 010: four points for $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$). ESA ★★★ ESA Dec-23 1c · ESA Jan-24 s1 1a

Why?

Why must f_x and f_y both vanish at an extremum?
Along the line $y = b$, $f(x, b)$ is a function of x alone with an extremum at $x = a$, so its derivative $f_x(a, b) = 0$. The same along $x = a$ gives $f_y(a, b) = 0$. Geometrically, the tangent plane is horizontal.

(added for clarity) T009 00:06:19

Why does $rt - s^2 < 0$ mean a saddle?
Near a stationary point $f(a+h, b+k) - f(a,b) \approx \frac{1}{2}(rh^2 + 2shk + tk^2)$ (Taylor, class 007). This quadratic keeps one sign for all h, k exactly when $rt - s^2 > 0$ (sign of r); when $rt - s^2 < 0$ it is positive in some directions and negative in others.

ADDED FOR CLARITY S009·6 · S009·7

Why is r enough to separate maximum from minimum when $rt - s^2 > 0$?
$rt > s^2 \geq 0$, so r and t have the same sign: the surface curves the same way along x and along y.

ADDED FOR CLARITY S009·8

Exam traps

Exam trap. $rt - s^2 < 0$ is a saddle point (a definite answer); only $rt - s^2 = 0$ is "no conclusion". The lecture mixes the two at one point. S009·7 · T009 00:08:23 · T009 00:11:06

Exam trap. Do not decide maximum or minimum from r before checking $rt - s^2 > 0$: $x^2 - 3xy + y^2$ has $r = 2 > 0$ at the origin but $rt - s^2 = -5$: a saddle. ADDED FOR CLARITY S009·8

Exam trap. $f_x = f_y = 0$ is necessary, $rt - s^2 > 0$ sufficient; slide 009·7 calls the pair "necessary and sufficient". ADDED FOR CLARITY S009·7

Exam trap. Evaluate r, s, t at each stationary point; a sign that holds at one point need not hold at another. S009·8

M9

Class 010 · Problems on Maxima and Minima

39 The method, and the ESA cubic functions

Definitions

Stationary (critical) points. The solutions of $f_x = 0$, $f_y = 0$, solved simultaneously; the points at which the function may have an extreme value. S010·7 · T010a 00:04:13 · T010b 00:01:23

Maximum / minimum value. The value $f(a, b)$ at a point of maximum or minimum, obtained by substituting the point into f. S010·8 · T010b 00:04:02

Statements in the class material

> *"If $rt - s^2 > 0$ and $r < 0$ at (a, b), $f(a, b)$ is a maximum value; if $rt - s^2 > 0$ and $r > 0$, a minimum value; if $rt - s^2 < 0$, $f(a, b)$ is not an extreme value, i.e. (a, b) is a saddle point; if $rt - s^2 = 0$ the case is doubtful and needs further investigation (class 009, restated)."* S010·8 · T010b 00:05:00

Key points

1. Procedure (lecture): find $p = f_x$, $q = f_y$; equate both to zero; find the critical points; compute r, s, t; evaluate $rt - s^2$ at each critical point; conclude. ESA ★★★ T010b 00:05:00 · T010b 00:05:16
2. Split $f_y = 0$ into cases and substitute each into $f_x = 0$: for the ESA cubic, $y = 0$ gives $x = 4, 6$ and $x = 5$ gives $y = \pm 1$. ESA ★★★ S010·7 · S010·8 · T010b 00:01:40
3. Count the stationary points before classifying; a cubic in x and y typically gives four, a quartic up to nine. ESA ★★ added for clarity S010·8 · S010·11
4. A saddle point has no value to report; give the maximum and minimum values only. ESA ★★ S010·8
5. When $rt - s^2 > 0$, r and t have the same sign, so either decides between maximum and minimum. ESA ★ added for clarity S010·8

Why?

Why does $f_y = 6y(x - 5) = 0$ give two separate cases?
A product is zero when either factor is zero: $y = 0$ or $x = 5$. Each case is substituted into $f_x = 0$ to find its own points; missing a case loses stationary points.

added for clarity S010·7 · T010b 00:01:40

Exam traps

Exam trap. Slide 010·8 says the maximum value is at $(6, 0)$; the maximum 112 is at $(4, 0)$, the minimum 108 at $(6, 0)$. S010·8 · T010b 00:04:42

Exam trap. $rt - s^2 = 36(x-5)^2 - 36y^2$: the 36 multiplies y^2 too (slide 010·8 drops it).

ADDED FOR CLARITY S010·8

40 Functions with many stationary points

Key points

1. Separable functions such as $x^4 + y^4 - x^2 - y^2 + 1$ give f_x in x only and f_y in y only: combine every root of one with every root of the other. ESA ★★ S010·11 · T010b 00:08:29
2. For $\sin x + \sin y + \sin(x + y)$, subtract the two conditions to get $\cos x = \cos y$, i.e. $x = y$ in the working range; then $\cos 2x = -\cos x = \cos(\pi - x)$ gives $x = \frac{\pi}{3}$. ESA ★★ S010·13 · T010b 00:10:23

Exam traps

Exam trap. When $rt - s^2 < 0$ the sign of r is irrelevant; at $(\pm\frac{1}{\sqrt{2}}, 0)$, $r = 4$ (not -2 as slide 010·12 says), and the point is still a saddle. ADDED FOR CLARITY S010·12 · T010b 00:09:26

Exam trap. At $(\frac{\pi}{3}, \frac{\pi}{3})$, $s = -\frac{\sqrt{3}}{2}$, so $rt - s^2 = 3 - \frac{3}{4} = \frac{9}{4}$ (slide 010·14 prints $\frac{3}{4}$).
ADDED FOR CLARITY S010·14

41 Applied problems: eliminate one variable first

Definitions

Elimination. Using the given condition (constraint) to express one variable in terms of the others, so that the quantity becomes a function of two independent variables. T010a 00:00:40 · T010a 00:08:27

Key points

1. We cannot yet find extrema of a function of three variables, so use the constraint to eliminate one (here z^2 or z). ESA ★★ T010a 00:00:40 · T010a 00:08:27
2. A product such as $kxyz^2$ is largest when x and y have the same sign and $k>0$; points with x, y of opposite signs give the lowest temperature. ESA ★★ S010·10 · T010b 00:05:40
3. Reject stationary points that make no physical sense (an edge $y=0$ in the box problem makes $x\to\infty$). ESA ★ T010a 00:13:07 · T010a 00:13:48
4. Elimination is not always easy; class 011's Lagrange multipliers avoid it (T010a 00:16:49). ESA ★ T010a 00:16:49

Why?

Why is a cube the wrong answer for an open box?
Without a top the base is used once and each side face twice in the area, so the box should be lower and wider: $x=y=\sqrt[3]{20}$, $z=\frac{1}{2}\sqrt[3]{20}$. The cube $x=y=z=\sqrt[3]{V}$ minimises the area of a closed box $2(xy+yz+zx)$.
ADDED FOR CLARITY S010·6 · CW U1 P34

Exam traps

Exam trap. $400xy(1-x^2-y^2)$ also has stationary points $(\pm1,0)$, $(0,\pm1)$ (saddles) that the slides omit; and slide 010·4's $rt-s^2$ values are -160000 and 320000, not -1600 and 356000.
ADDED FOR CLARITY S010·3 · S010·4

Exam trap. Open box: area $xy+2yz+2zx$ (one base, four sides). The slide's $2xy+2yz+zx$ and its derivative slip lead to a cube, which is wrong. ADDED FOR CLARITY S010·5 · S010·6 · T010a 00:07:38

M10

Class 011 · Lagrange's Method of Undetermined Multipliers

42 Lagrange's method: the auxiliary function and the procedure

Definitions

Constraint. A condition $\phi(x, y, z) = 0$ that the variables must satisfy (walk only on the surface of a sphere, a fixed surface area, a fixed sum). S011·3 · T011a 00:00:36

Lagrange multiplier. Suppose we need to minimize or maximize a function $f(x, y, z)$ subject to the constraint $\phi(x, y, z)$; then we can introduce an additional unknown constant λ, known as Lagrange's multiplier, to ease the process of finding the extrema. S011·3 · T011a 00:00:53

Auxiliary function. $F(x, y, z) = f(x, y, z) + \lambda\phi(x, y, z)$: the given function plus λ times its constraint (also called Lagrange's function). S011·3 · T011a 00:01:08 · T011b 00:02:06

Statements in the class material

> *"Working procedure. Step 1: Form the auxiliary equation $F(x, y, z) = f(x, y, z) + \lambda\phi(x, y, z)$ ... (1). Step 2: Partially differentiate F in (1) w.r.t. x, y, z respectively. Step 3: Solve the four equations $F_x = 0$, $F_y = 0$, $F_z = 0$ and the constraint for the Lagrange multiplier λ and stationary values x, y, z."* S011·3

> *"Advantages: 1. The stationary points of $f(x, y, z)$ can be determined without determining x, y, z explicitly. 2. This method can be extended to function of several variables subject to many constraints. Disadvantages: 1. Nature of the stationary points can not be determined. 2. Equations $F_x = 0$, $F_y = 0$, $F_z = 0$ are only necessary conditions."* S011·4 · T011a 00:03:04

Key points

1. The method avoids elimination: in class 010 we had to substitute $z^2 = 1 - x^2 - y^2$ for the temperature problem; with Lagrange we differentiate F directly. ESA ★★ T011a 00:01:18 · T011a 00:03:17
2. Four unknowns x, y, z, λ need four equations: $F_x = 0$, $F_y = 0$, $F_z = 0$ and the constraint $\phi = 0$. ESA ★★★ T011a 00:02:30
3. λ is "undetermined": we may find it, but it does not appear in the answer. ESA ★ T011a 00:02:04

4. The method gives only stationary points; to verify maximum or minimum by $rt - s^2$ we would have to go back to the previous method. In problems the question itself says which (maximum volume, closest point). ESA ★★ S011·4 · T011a 00:04:05 · T011a 00:14:01
5. The most important step is to identify the function f and the constraint ϕ correctly (Figure F42.2). ESA ★★★ T011a 00:09:15 · T011b 00:01:46
6. Geometrically, at a constrained extremum a level curve (surface) of f just touches the constraint, so ∇f and $\nabla \phi$ are parallel: $\nabla f = -\lambda \nabla \phi$ (Figure G42.1). ESA ★ ADDED FOR CLARITY S011·7

Why?

Why does $F_x = F_y = F_z = 0$ find the constrained extrema?
At a constrained extremum, f cannot increase along any direction that stays on the surface $\phi = 0$, so ∇f is perpendicular to the surface, i.e. parallel to $\nabla \phi$: $\nabla f + \lambda \nabla \phi = 0$, which is exactly $F_x = F_y = F_z = 0$.
ADDED FOR CLARITY S011·3 · S011·7

Exam traps

Exam trap. Do not forget the constraint as the fourth equation; three equations cannot fix four unknowns. T011a 00:02:30

Exam trap. Dividing by x, y, z or λ assumes it is not zero; check those cases separately (they usually give zero volume or an impossible point). ADDED FOR CLARITY S011·9 · T011b 00:03:40

43 Geometric problems: inscribed solids, nearest and farthest points

Definitions

Continued product. The product obtained by multiplying together in succession: for parts x, y, z, the product of the first, the square of the second and the cube of the third is xy^2z^3. S011·12 · T011b 00:05:23

Key points

1. For a box inscribed in a sphere or ellipsoid centred at the origin, take the corner (x, y, z) in the first octant: edges $2x, 2y, 2z$, volume $8xyz$. Using xyz instead gives the same point. ESA ★★★ S011·5 · T011a 00:06:08 · T011a 00:06:53
2. To find the nearest point, imagine a small sphere centred at the origin expanding like a soap bubble until it just touches the surface (Figure G43.1). ESA ★ S011·7
3. Reject solutions that are impossible: on $x^2 - z^2 = 1$, $x = 0$ would need $z^2 = -1$. ESA ★★ S011·9 · T011a 00:12:26
4. When there are two stationary values (W43.4: distances 1 and 2), the larger is the maximum and the smaller the minimum on a closed curve. ESA ★★★ ADDED FOR CLARITY ESA Jul-23 1c

Why?

Why minimise $x^2 + y^2 + z^2$ instead of the distance?
$\sqrt{u}$ increases with u, so both have their extrema at the same points, and the square has no square root to differentiate.

ADDED FOR CLARITY S011·8 · T011a 00:10:08

Exam traps

Exam trap. The box in the sphere has edges $2x, 2y, 2z$, so $V = 8xyz$: its largest volume is $\frac{8r^3}{3\sqrt{3}}$, not $\frac{r^3}{3\sqrt{3}}$. ADDED FOR CLARITY S011·5

Exam trap. Slide 011·6's third line should read $8xyz + 2z^2\lambda = 0$ (printed with y^2).

ADDED FOR CLARITY S011·6

44 Applied problems: boxes, splitting a number, temperature

Key points

1. Open box (no top): surface $xy + 2yz + 2zx$; closed box: $2(xy + yz + zx)$. ESA ★★ S011·10 · T011b 00:01:09
2. Dividing a number N into parts x, y, z: constraint $x + y + z = N$; the function is the product asked for. ESA ★ S011·12 · T011b 00:05:57
3. The temperature problem of class 010 gives the same 50 by Lagrange's method with no elimination (W44.3). ESA ★★★ T011a 00:14:33 · ESA Jun-24 1d

Exam traps

Exam trap. Open-top box: $x = y = 2z$ (height half the base edge), not a cube. S011·11 · CW U1 P34

★ Past-paper questions

Class 001. Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. No question asks for a limit, continuity or the definition of a partial derivative. Computing partial derivatives is asked directly twice: Dec 2023 Q1.a (third-order partials of $e^x \log y + \cos y \log x$) and Jan 2024 set 4 Q1.c (find n for $\theta = t^n e^{-r^2/4t}$). Both are worked in the class 002 pack; this pack works the class-work problems of the same type (CW P4, P7, P8). Every other Unit 1 question (chain rule, Euler, Taylor, maxima and minima, Lagrange) uses the rules of this session.

Class 002. Eight past ESA papers were checked: UE22MA141A Dec 2023, May 2024 and Jul 2023, and UE23MA141A Jan sets 1–4 and June 2024. None asks for the geometrical interpretation directly. Two 6-mark Q1 parts test higher-order partials, the computational half of this session. Other Unit 1 questions (Euler, Taylor, maxima, Lagrange, chain rule) belong to later sessions and are left out.

Classes 003–004. Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. In-scope Q1 parts: five chain-rule questions (Jan 2024 set 1 Q1.d and set 2 Q1.d, 5 marks each; set 3 Q1.b, set 4 Q1.b and June 2024 Q1.a, 6 marks each) and one implicit-function question (Jan 2024 set 3 Q1.d, 6 marks). The total derivative and rates of change were not asked in these papers; they are computed in class-work problems 9–13. No paper repeats another. Questions on Euler's theorem, Taylor series, maxima and Lagrange belong to later sessions and are left out.

Class 005. Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. Euler's theorem appears in six of them: Jul 2023 Q1.a, May 2024 Q1.a, June 2024 Q1.b (the same question as May 2024), Jan 2024 set 3 Q1.c, set 4 Q1.d, and set 1 Q1.d (a degree-0 function, also solvable by the chain rule). Five are 6-mark questions, so this is one of the most examined topics in Unit 1. The definition, applications and third-order result were not asked.

Class 006. Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. Euler problems appear in six of them: Jul 2023 Q1.a (degree 5: $5u$, $20u$), May 2024 Q1.a and June 2024 Q1.b (the same $\tan^{-1}$ extension question), Jan 2024 set 3 Q1.c ($17\pi/2$), set 4 Q1.d ($9u$) and set 1 Q1.d (degree 0). Five are 6-mark questions. Testing homogeneity alone and verifying Euler's theorem by direct differentiation (topics 1–2) were not asked on their own; they appear only as the first step of a longer question.

Class 007. Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. A Taylor or Maclaurin expansion in two variables is asked in six of them, always as a 6–7 mark Q1 part. About a point (a, b): Dec 2023 Q1.b ($\sin x \sin y$, third degree), Jul 2023 Q1.b (the same function, second order) and May 2024 Q1.b ($e^x \cos y$ about $(1, \pi/4)$). About the origin:

Jan 2024 set 1 Q1.b ($\log(1+x-y)$), set 2 Q1.b ($e^{ax}\sin by$) and June 2024 Q1.c ($\sin(x+2y)$). Four of the six ask for third-degree terms, which the slides do not write out. Jan 2024 sets 3 and 4 have no Taylor question. One-variable series and the applications were never asked.

Class 008. Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. A two-variable Taylor or Maclaurin expansion is asked in six of them, always a 6–7 mark Q1 part: Jul 2023 Q1.b and Dec 2023 Q1.b ($\sin x \sin y$ about $(\frac{\pi}{4}, \frac{\pi}{4})$, second order and third degree), May 2024 Q1.b ($e^x \cos y$ about $(1, \frac{\pi}{4})$, the same as slide 008·3), Jan 2024 set 1 Q1.b ($\log(1+x-y)$), set 2 Q1.b ($e^{ax}\sin by$) and June 2024 Q1.c ($\sin(x+2y)$). Four ask for third-degree terms. Jan 2024 sets 3 and 4 have no Taylor question. Approximate values (topic 3) were never asked in these papers; they appear in class and in CW problem 26.

Class 009. Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. The definitions are never asked by themselves, but the test of this session is the whole of three 7-mark Q1 parts: Dec 2023 Q1.c and Jan 2024 set 1 Q1.a (the same function $x^3+3xy^2-15x^2-15y^2+72x$) and Jan 2024 set 2 Q1.a ($x^3-12x+y^3+3y^2-9y$). They are worked in the class 010 pack; the method is starred ★★★ here.

Class 010. Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. Free extrema of $f(x,y)$ are asked in three of them, always a 7-mark Q1 part: Dec 2023 Q1.c and Jan 2024 set 1 Q1.a ask the same function $x^3+3xy^2-15x^2-15y^2+72x$ (also slide 010·7 and CW problem 32), and Jan 2024 set 2 Q1.a asks $x^3-12x+y^3+3y^2-9y$ (CW problem 29). Both are cubic polynomials with four stationary points: one maximum, one minimum and two saddle points. The other five papers ask Lagrange's method (class 011) instead. The temperature-on-a-sphere problem of slide 010·3 is asked in June 2024 Q1.d, but by Lagrange's method; the elimination method of topic 3 gives the same answer. Trigonometric and quartic functions (topic 2) were never asked.

Class 011. Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. Lagrange's method is asked in five of them, always as a 7-mark Q1 part, with four distinct problems: Jul 2023 Q1.c (maximum and minimum distances from the origin to $5x^2+6xy+5y^2-8=0$), May 2024 Q1.c (the rectangular solid of maximum volume in a sphere is a cube; also slides 011·5–6), Jan 2024 sets 3 and 4 Q1.a (largest parallelepiped in an ellipsoid) and June 2024 Q1.d (highest temperature $400xyz^2$ on the unit sphere; also class 010 by elimination). Three of the four also appear in the extra question list (Q57, Q59, Q61). Dec 2023 and Jan 2024 sets 1–2 ask free extrema (class 010) instead.

12 · Problems on partial derivatives

ESA Dec 2023 · 6 marks · UE22MA141A Q1.a · solved in W12.4

If $f(x,y)=e^x\log y+\cos y\log x$, find f_{xxx}, f_{xyy} and f_{yyy} at $(1,\pi/2)$.

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q1.c · solved in W12.5

If $\theta = t^n e^{-r^2/4t}$, what value of n will make $\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial \theta}{\partial r}\right) = \frac{\partial \theta}{\partial t}$?

17 · The chain rule for composite functions

ESA Jan 2024 · 5 marks · UE23MA141A set 1, Q1.d · solved in W17.11

If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, find the value of $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$.

ESA Jan 2024 · 5 marks · UE23MA141A set 2, Q1.d · solved in W17.12

If $u = f(x-y,\, y-z,\, z-x)$, find the value of $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$.

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q1.b · solved in W17.13

If $z = f(u, v)$ and $u = x^2 - y^2$, $v = 2xy$, show that $\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = 4(u^2+v^2)^{1/2}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right]$.

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q1.b · solved in W17.14

If $z = f(x, y)$ and $x = e^u \sin v$, $y = e^u \cos v$, show that $\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = e^{-2u}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right]$.

ESA June 2024 · 6 marks · UE23MA141A Q1.a · solved in W17.15

If $z = \log(u^2 + v)$, where $u = e^{x+y^2}$, $v = x + y^2$, then find the value of $2y\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}$.

19 · Implicit functions of three variables

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q1.d · solved in W19.3

For the point on the surface $x^x y^y z^z = c$, show that at $x = y = z$, $\frac{\partial^2 z}{\partial x \partial y} = -[x\log(ex)]^{-1}$.

22 · Second and higher order: the corollary

ESA Jul 2023 · 6 marks · UE22MA141A Q1.a · solved in W22.1

If $u(x, y) = x^3 y^2 \sin^{-1}\left(\frac{y}{x}\right)$, prove that (i) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 5u$, (ii) $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 20u$.

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q1.c · solved in W22.5

If $u = x^3 \sin^{-1}\left(\frac{y}{x}\right) + x^4 \tan^{-1}\left(\frac{y}{x}\right)$, find $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$ at $x = 1$, $y = 1$ by using Euler's theorem and its extension.

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q1.d · solved in W22.4

If $u = x^3 \sin^{-1}\left(\frac{y}{x}\right) + y^{-3}\tan^{-1}\left(\frac{x}{y}\right)$, evaluate $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$ by using Euler's theorem and its extension.

23 · Extension: when f(u) is homogeneous

ESA May 2024 · 6 marks · UE22MA141A Q1.a · solved in W23.1

If $u = \tan^{-1}\left(\frac{x^4+y^4}{xy}\right)$, prove that (i) $xu_x + yu_y = \sin 2u$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \sin 2u(1 - 4\sin^2 u)$.

ESA June 2024 · 6 marks · UE23MA141A Q1.b · solved in W23.1

Same question as May 2024 Q1.a: $u = \tan^{-1}\left(\frac{x^4+y^4}{xy}\right)$, prove (i) and (ii).

24 · Applications of homogeneous functions

ESA Jan 2024 · 5 marks · UE23MA141A set 1, Q1.d · solved in W24.1

If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, find $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$ (degree 0; also solvable by the chain rule).

27 · When u itself is homogeneous: first and second order

ESA Jul 2023 · 6 marks · UE22MA141A Q1.a · solved in W27.4

If $u(x, y) = x^3y^2 \sin^{-1}\left(\frac{y}{x}\right)$, prove that (i) $xu_x + yu_y = 5u$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = 20u$.

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q1.c · solved in W27.7

If $u = x^3 \sin^{-1}\left(\frac{y}{x}\right) + x^4\tan^{-1}\left(\frac{y}{x}\right)$, find $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$ at $x = 1$, $y = 1$ by using Euler's theorem and its extension.

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q1.d · solved in W27.6

If $u = x^3 \sin^{-1}\left(\frac{y}{x}\right) + y^{-3}\tan^{-1}\left(\frac{x}{y}\right)$, evaluate $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$ by using Euler's theorem and its extension.

ESA Jan 2024 · 5 marks · UE23MA141A set 1, Q1.d · solved in W27.8

If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, find $xu_x + yu_y + zu_z$ (degree 0; also solvable by the chain rule).

28 · When f(u) is homogeneous: the extension

ESA May 2024 · 6 marks · UE22MA141A Q1.a · solved in W28.5

If $u = \tan^{-1}\left(\frac{x^4+y^4}{xy}\right)$, prove that (i) $xu_x + yu_y = \sin 2u$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \sin 2u(1 - 4\sin^2 u)$.

ESA June 2024 (6 marks) UE23MA141A Q1.b · solved in W28.5

Same question as May 2024 Q1.a: $u = \tan^{-1}\left(\frac{x^4+y^4}{xy}\right)$, prove (i) and (ii).

31 · Taylor's expansion of a function of two variables

ESA Jul 2023 (7 marks) UE22MA141A Q1.b · solved in W31.1

Expand $f(x,y) = \sin x \sin y$ in Taylor's series up to second order terms about the point $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$.

ESA Dec 2023 (7 marks) UE22MA141A Q1.b · solved in W31.2

Expand $f(x,y) = \sin x \sin y$ in Taylor's series up to third degree term about $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$.

ESA May 2024 (7 marks) UE22MA141A Q1.b · solved in W31.3

Expand $f(x,y) = e^x \cos y$ in powers of $(x-1)$ and $\left(y - \frac{\pi}{4}\right)$ using Taylor's series up to second degree terms.

32 · Maclaurin's expansion of a function of two variables

ESA Jan 2024 (7 marks) UE23MA141A set 1, Q1.b · solved in W32.1

Expand $\log(1 + x - y)$ about origin up to third degree terms.

ESA Jan 2024 (7 marks) UE23MA141A set 2, Q1.b · solved in W32.2

Expand $e^{ax} \sin by$ about origin up to third degree terms.

ESA June 2024 (6 marks) UE23MA141A Q1.c · solved in W32.3

Expand $f(x,y) = \sin(x + 2y)$ in Taylor's series up to third degree terms about the point $(0,0)$.

33 · Taylor expansion about a point (a, b)

ESA May 2024 (7 marks) UE22MA141A Q1.b · solved in W33.1

Expand $f(x,y) = e^x \cos y$ in powers of $(x-1)$ and $\left(y - \frac{\pi}{4}\right)$ using Taylor's series up to second degree terms.

ESA Jul 2023 (7 marks) UE22MA141A Q1.b · solved in W33.4

Expand $f(x,y) = \sin x \sin y$ in Taylor's series up to second order terms about the point $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$.

ESA Dec 2023 (7 marks) UE22MA141A Q1.b · solved in W33.5

Expand $f(x,y) = \sin x \sin y$ in Taylor's series up to third degree term about $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$.

34 · Maclaurin expansion about the origin

ESA Jan 2024 · 7 marks · UE23MA141A set 1, Q1.b · solved in W34.4

Expand $\log(1 + x - y)$ about origin up to third degree terms.

ESA Jan 2024 · 7 marks · UE23MA141A set 2, Q1.b · solved in W34.5

Expand $e^{ax} \sin by$ about origin up to third degree terms.

ESA June 2024 · 6 marks · UE23MA141A Q1.c · solved in W34.6

Expand $f(x, y) = \sin(x + 2y)$ in Taylor's series up to third degree terms about the point $(0, 0)$.

39 · The method, and the ESA cubic functions

ESA Dec 2023 · 7 marks · UE22MA141A Q1.c · solved in W39.1

Find the extreme values of the function $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$.

ESA Jan 2024 · 7 marks · UE23MA141A set 1, Q1.a · solved in W39.1

Locate the stationary points of $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$. Discuss their nature.

ESA Jan 2024 · 7 marks · UE23MA141A set 2, Q1.a · solved in W39.2

Locate the stationary points of $f(x, y) = x^3 - 12x + y^3 + 3y^2 - 9y$. Discuss their nature.

41 · Applied problems: eliminate one variable first

ESA June 2024 · 7 marks · UE23MA141A Q1.d · solved in W41.1

The temperature T at any point (x, y, z) in space is $400xyz^2$. Using Lagrange's method of multipliers, find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$. (Asked by Lagrange's method; W41.1 checks the answer by elimination.)

43 · Geometric problems: inscribed solids, nearest and farthest points

ESA Jul 2023 · 7 marks · UE22MA141A Q1.c · solved in W43.4

Find the maximum and minimum distances from the origin to the curve $5x^2 + 6xy + 5y^2 - 8 = 0$.

ESA May 2024 · 7 marks · UE22MA141A Q1.c · solved in W43.1

Show that the rectangular solid of maximum volume that can be inscribed in a given sphere is a cube by using Lagrange's method of multipliers.

ESA Jan 2024 7 marks UE23MA141A set 3, Q1.a · solved in W43.2

Use the method of Lagrange's multipliers to find the volume of the largest rectangular parallelepiped that can be inscribed in the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$.

ESA Jan 2024 7 marks UE23MA141A set 4, Q1.a · solved in W43.2

The same question as set 3 Q1.a (largest parallelepiped in the ellipsoid).

44 · Applied problems: boxes, splitting a number, temperature

ESA June 2024 7 marks UE23MA141A Q1.d · solved in W44.3

The temperature T at any point (x, y, z) in space is $400xyz^2$. Using Lagrange's method of multipliers, find the highest temperature on the surface of the unit sphere $x^2+y^2+z^2=1$.